

NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

GROUP 3 SC2006 - SCSD

LAB 4 DELIVERABLES

Team Members
1. Vivek Shrey (U2440609F) - Team Leader
2. Ching Xin Wei (U2421032K)
3. Dharshini Saravanan (U2420758E)
4. Li Sinuo (U2421736H)
5. Sivasankar Sivagnana Sundari (U2420528F)

I. Black Box Testing

A. AuthenticationController

Control Class to test - **AuthenticationController**

The AuthenticationController class is responsible for managing user authentication within the application, encompassing both user registration and user login functionalities.

The AuthenticationController is an integral component that facilitates secure interaction between the client and the authentication services of the system. When a user attempts to log in, the controller receives their login credentials, specifically the email and password. It then forwards these details to the AuthService for authentication. The system validates the provided credentials to ensure they match the stored records within the database. If the verification process confirms that the email exists and the password hash corresponds correctly to the stored hashed value, the login is deemed successful, and the user is granted access to the application. In this process, the controller also retrieves user information such as user ID, name, role, and associated skills to return a structured response to the client. The primary information managed by the AuthenticationController during user login includes the user's email, password hash, and related account details retrieved upon successful authentication.

In addition to login functionality, the AuthenticationController also manages user registration, which allows new users to create an account within the system. During registration, the controller collects essential information such as name, email, password, role, and a list of skills. The collected data is validated to ensure accuracy and completeness before being processed. The password provided by the user undergoes a secure hashing procedure prior to storage to maintain confidentiality and prevent unauthorized access. After validation, the controller communicates with the AuthService to store the new user's details in the database. If the user has provided a list of skills, these are subsequently added and linked to the registered account.

The AuthenticationController also handles special conditions within registration, such as checking for duplicate email entries or invalid input formats. In such cases, it returns appropriate error messages indicating the nature of the registration failure. Upon successful registration, a unique user ID is generated and assigned to the new user. This identifier becomes crucial for tracking user data and managing authentication across the application. Information managed by the AuthenticationController in the registration process includes name, email, password, role, school, graduation year, and skills.

B. Equivalence Class and Boundary Value Testing

Equivalence Class Testing

Equivalence Class Testing divides input data into valid and invalid partitions (or classes) that are expected to behave similarly. The technique aims to identify subsets of data that are logically equivalent in how the program processes them, allowing testers to select representative values from each class instead of testing every possible input. This partitioning is typically done based on the system's input specifications, constraints, and validation rules. Each equivalence class represents a range or set of conditions that share common characteristics, such as format, type, or range validity.

For instance, valid equivalence classes represent input data that meet all functional requirements and constraints of the system, while invalid equivalence classes contain data that violate one or more of these constraints. The division ensures that the testing process identifies potential defects arising from incorrect handling of valid inputs as well as inadequate rejection of invalid inputs.

By testing one representative from each class, redundancy is reduced while ensuring broad coverage of input possibilities.

Boundary Value Testing

Boundary Value Testing focuses on testing input values at the edges of allowed ranges. It helps detect off-by-one and input validation errors. This ensures the system behaves correctly under extreme but valid conditions, and fails gracefully when limits are exceeded.

1. Login function

Valid Equivalence Class: Emails and passwords with correct formats.
Invalid Equivalence Class: Emails and passwords with incorrect formats or missing information.

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2. Register function

Valid Equivalence Class: Name, Email, Password, Role, Skills with correct formats.
Invalid Equivalence Class: Name, Email, Password, Role, Skills with incorrect formats or missing information.

C. Test Cases and Results

a) Login

Input parameters: Email and password

No.	Test Input	Expected Output	Actual Output	Pass?
1	**(Correct email and password)** **(Valid)** Email: “test123@gmail.com” **(Valid)** Password: “Password123”	Successful login	Successful login	YES
2	**(Valid email but wrong password)** **(Valid)** Email: “test123@gmail.com” **(Invalid)** Password: “Pass222222”	Login failed, system notify “Something went wrong during login. Please try again later.”	Login failed, system notify “Something went wrong during login. Please try again later.”	YES
3	**(Invalid email format)** **(Invalid)** Email: “test123” **(Valid)** Password: “Password123”	Login failed, system notify “Please enter a valid email address. Please fix the highlighted fields and try again”	Login failed, system notify “Please enter a valid email address. Please fix the highlighted fields and try again”	YES
4	**(Empty string for email)** **(Invalid)** Email: “” **(Valid)** Password: “Password123”	Login failed, system notify “Email is required. Please fix the highlighted fields and try again”	Login failed, system notify “Email is required. Please fix the highlighted fields and try again”	YES
5	**(Empty password)** **(Valid)** Email: “test123@gmail.com” **(Invalid)** Password: “”	Login failed, system notify “Password is required. Please fix the highlighted fields and try again”	Login failed, system notify “Password is required. Please fix the highlighted fields and try again”	YES

b) Register

Input parameters: Name, Email, Password, Role, Skills

No.	Test Input	Expected Output	Actual Output	Pass?
1	**(All required fields provided correctly)** Name: “John Tester” Email: “test123@gmail.com” Password: “Password123” Role: “Fresh Graduate”	Successful registration	Successful registration	YES
2	**(Missing name)** Name: ””	Registration failed, system notify “Name is required”	Registration failed, system notify “Name is required”	YES
3	**(Missing email)** Email: “”	Registration failed, system notify “Please enter a valid email address”	Registration failed, system notify “Please enter a valid email address”	YES
4	**(Invalid email format)** Email: “test123”	Registration failed, system notify “Please enter a valid email address”	Registration failed, system notify “Please enter a valid email address”	YES
5	**(Missing password)** Password:””	Registration failed, system notify “Password must be at least 8 characters long”	Registration failed, system notify “Password must be at least 8 characters long”	YES
6	**(Invalid password format)** Password: “pass”	Registration failed, system notify “Password must be at least 8 characters long”	Registration failed, system notify “Password must be at least 8 characters long”	YES
7	**(Invalid password format)** Password: “password”	Registration failed, system notify “Password must contain at least one uppercase letter”	Registration failed, system notify “Password must contain at least one uppercase letter”	YES
8	**(Duplicate email registration)**	Registration failed, system notify “Registration failed: Email already registered”	Registration failed, system notify “Registration failed: Email already registered”	YES
9	**(Missing optional fields)** Role:””	Registration failed, system notify	Registration failed, system notify	YES

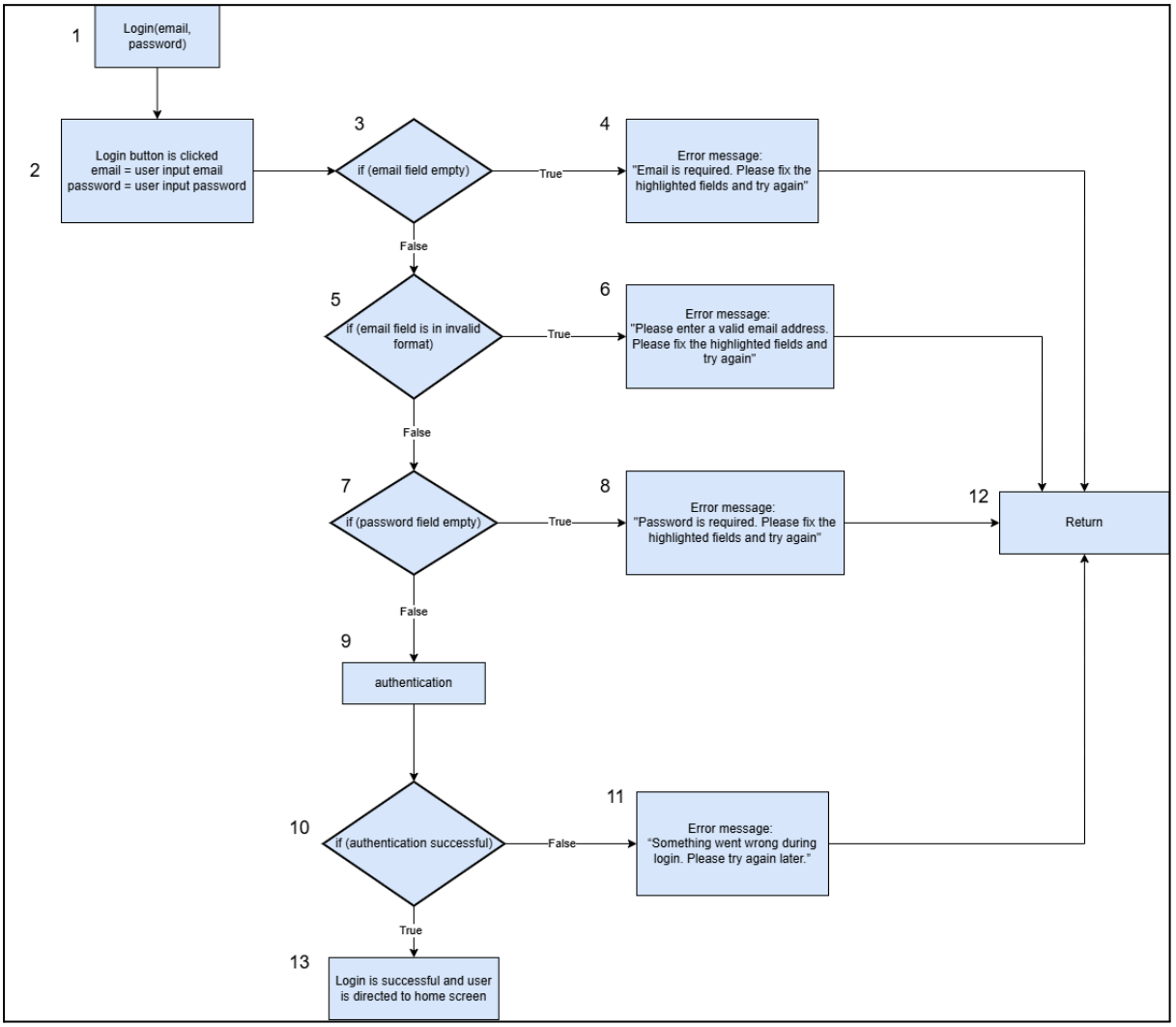
		“Please select your current role”	“Please select your current role”	
10	**(Less than 2 skills selected)**	Registration failed, system notify “Please select at least 2 skills”	Registration failed, system notify “Please select at least 2 skills”	YES

II. White Box Testing

A. Login

Control Flow Graph

If the image is unclear, please refer to the separate file on GitHub.



Basis Path Testing

Cyclomatic Complexity: |decision points| + 1 = 4 + 1 = 5

Basis Paths:

1. Baseline path: 1, 2, 3, 5, 7, 9, 10, 13
2. Basis path #2: 1, 2, 3, 4, 12
3. Basis path #3: 1, 2, 3, 5, 6, 12
4. Basis path #4: 1, 2, 3, 5, 7, 8, 12
5. Basis path #5: 1, 2, 3, 5, 7, 9, 10, 11, 12

Test Cases and Results

Login(email, password)

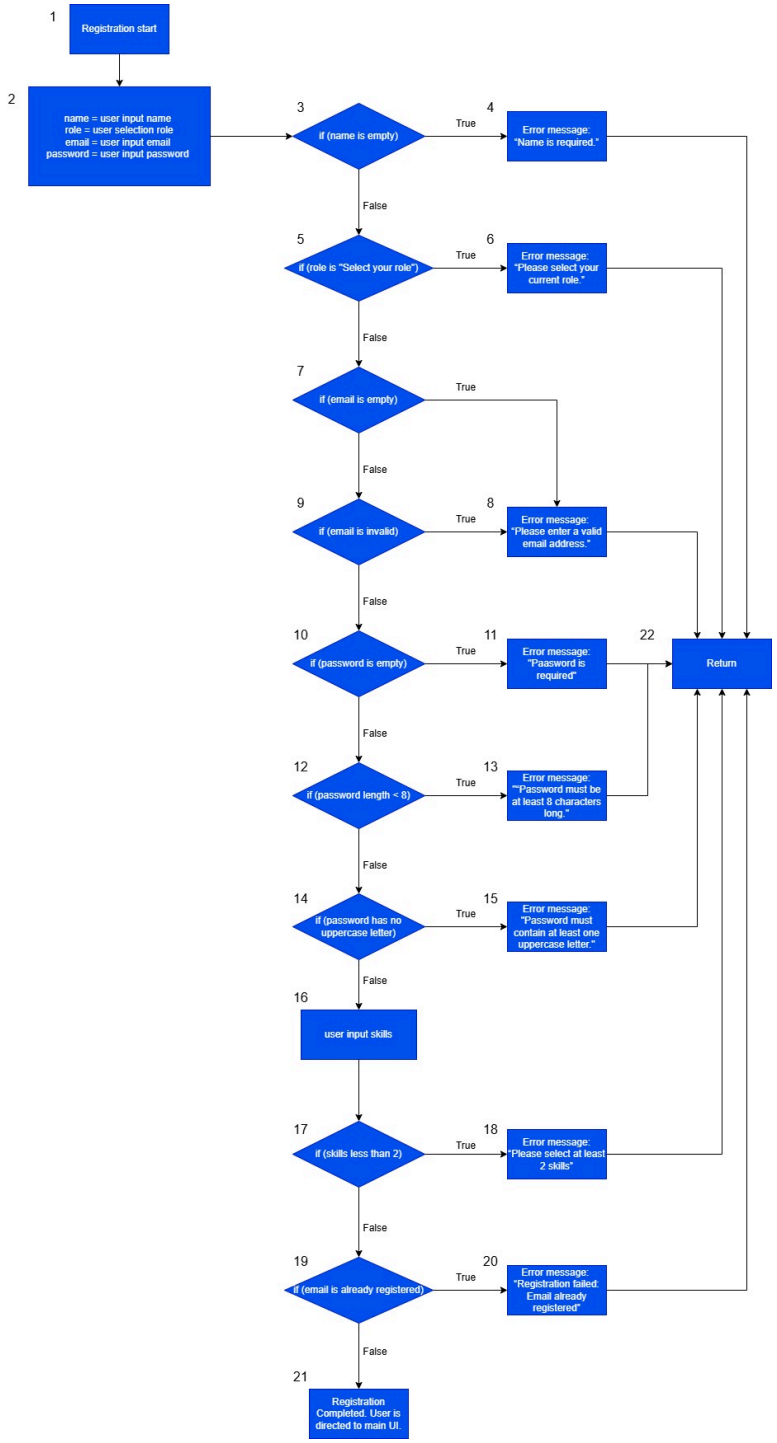
No.	Test Input	Expected Output	Actual Output	Pass?
1	Email = “test123@gmail.com” Password = “Password123”	Successful login	Successful login	YES
2	Email = “” Password = “Password123”	Login failed, system notify “Email is required. Please fix the highlighted fields and try again”	Login failed, system notify “Email is required. Please fix the highlighted fields and try again”	YES
3	Email = “test123” Password = “Password123”	Login failed, system notify “Please enter a valid email address. Please fix the highlighted fields and try again”	Login failed, system notify “Please enter a valid email address. Please fix the highlighted fields and try again”	YES
4	Email = “test123@” Password = “Password123”	Login failed, system notify “Please enter a valid email address. Please fix the highlighted fields and try again”	Login failed, system notify “Please enter a valid email address. Please fix the highlighted fields and try again”	YES
5	Email = “test123@gmail.com” Password = “”	Login failed, system notify “Password is required. Please fix the highlighted fields and try again”	Login failed, system notify “Password is required. Please fix the highlighted fields and try again”	YES
6	Email = “test123@gmail.com” Password = “WrongPass222”	Login failed, system notified “Something went wrong during login. Please try again later.”	Login failed, system notified “Something went wrong during login. Please try again later.”	YES
7	Email = “wrong123@gmail.com”	Login failed, system notified “Something	Login failed, system notified “Something	YES

	Password = “Password123”	went wrong during login. Please try again later.”	went wrong during login. Please try again later.”	

B. Registration

Control Flow Graph

If the image is unclear, please refer to the separate file on GitHub.



Basis Path Testing

Cyclomatic Complexity: |decision points| + 1 = 9 + 1 = 10

Basis Paths:

1. Baseline path: 1, 2, 3, 5, 7, 9, 10, 12, 14, 15, 17, 19, 21
2. Basis path #2: 1, 2, 3, 4, 22
3. Basis path #3: 1, 2, 3, 5, 6, 22
4. Basis path #4: 1, 2, 3, 5, 7, 8, 22
5. Basis path #5: 1, 2, 3, 5, 7, 9, 8, 22
6. Basis path #6: 1, 2, 3, 5, 7, 9, 10, 11, 22
7. Basis path #7: 1, 2, 3, 5, 7, 9, 10, 12, 13, 22
8. Basis path #8: 1, 2, 3, 5, 7, 9, 10, 12, 14, 15, 22
9. Basis path #9: 1, 2, 3, 5, 7, 9, 10, 12, 14, 15, 17, 18, 22
10. Basis path #10: 1, 2, 3, 5, 7, 9, 10, 12, 14, 15, 17, 19, 20, 22

Test Cases and Results

No.	Test Input	Expected Output	Actual Output	Pass?
1	name="John", email="john@gmail.com", passwordHash="Password123", role="Fresh Graduate", skills=["Java","Python"]	Registration successful, system displays main UI.	Registration successful, system displayed main UI.	YES
2	name="", email="jane@gmail.com", password="Password123", role="Fresh Graduate", skills=["Java”, “Python"]	Exception raised, system notifies “Name is required.”	Registration failed, system notified “Name is required.” Cannot proceed to the skill selection page.	YES
3	name="Jane", email="jane@gmail.com", passwordHash="Password123", role=”Select your role”, skills=["Java”, “Python"]	Exception raised, system notifies “Please select your current role.”	Registration failed, system notified “Please select your current role.” Cannot proceed to the skill selection page.	YES
4	name="Jane", email="", password="Password123", role="Fresh Graduate", skills=["Java”, “Python"]	Exception raised, system notifies “Please enter a valid email address.”	Registration failed, system notified “Please enter a valid email address.” Cannot proceed to the skill selection page.	YES
5	name="Jane",	Exception raised,	Registration failed,	YES

	email="jane@g", password="Password123", role="Fresh Graduate", skills=["Java”, “Python"]	system notifies “Please enter a valid email address.”	system notified “Please enter a valid email address.” Cannot proceed to the skill selection page.	
6	name="Jane", email="john@gmail.com", password="Password123", role="Fresh Graduate", skills=["Java”, “Python"]	Exception raised, system notifies “Registration failed: Email already registered.”	Registration failed, system notified “Registration failed: Email already registered” after skill selection.	YES
7	name="Jane", email="jane@gmail.com", password="", role=”Fresh Graduate”, skills=["Java”, “Python"]	Exception raised, system notifies “Password is required.”	Registration failed, system notified “Password is required.” Cannot proceed to the skill selection page.	YES
8	name="Jane", email="jane@gmail.com", password="123", role=”Fresh Graduate”, skills=["Java”, “Python"]	Exception raised, system notifies “Password must be at least 8 characters long.”	Registration failed, system notified “Password must be at least 8 characters long.” Cannot proceed to the skill selection page.	YES
9	name="Jane", email="jane@gmail.com", password="12345678", role=”Fresh Graduate”, skills=["Java”, “Python"]	Exception raised, system notifies “Password must contain at least one uppercase letter.”	Registration failed, system notified “Password must contain at least one uppercase letter.” Cannot proceed to the skill selection page.	YES
10	name="Jane", email="jane@gmail.com", password="Password123", role=”Fresh Graduate”, skills=[]	Exception raised, system notifies “Please select at least 2 skills.”	Registration failed, system notified “Please select at least 2 skills” at the skill selection page.	YES